Horizon Project

Generated by Doxygen 1.9.8

Chapter 1

Namespace Index

1.1 Package List

Here are the packages with brief descriptions (if available):

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

Chapter 5

Namespace Documentation

5.1 horizon Namespace Reference

Namespaces

- namespace fd_pattern_graph
- namespace fds
- namespace horizon
- namespace static_fd_analysis
- namespace utils

5.2 horizon.fd_pattern_graph Namespace Reference

Classes

- class FDPatternGraph

Variables

- logger = get_logger(__name__)

5.2.1 Detailed Description

```
FDPatternGraph: Build a NetworkX graph from CSV data with Functional Dependency edges.

Each unique (column, value) pair becomes a node. This means:
- Same values in the same column share the same node
- Same values in different columns have different nodes
- Edges represent functional dependencies between column-value pairs
```

5.2.2 Variable Documentation

5.2.2.1 logger

```
horizon.fd_pattern_graph.logger = get_logger(__name__)
```

5.3 horizon.fds Namespace Reference

Namespaces

- namespace fd
- namespace fd_pattern
- namespace pattern_expression
- namespace set_of_fds

5.3.1 Detailed Description

```
Classes relating to functional dependencies.
```

5.4 horizon.fds.fd Namespace Reference

Classes

- class FunctionalDependency

5.4.1 Detailed Description

```
FunctionalDependency: Representing a functional dependency with one value on the RHS and potentially multiple
```

5.5 horizon.fds.fd_pattern Namespace Reference

Classes

- class FDPattern

5.6 horizon.fds.pattern_expression Namespace Reference

Classes

- class PatternExpression

5.7 horizon.fds.set_of_fds Namespace Reference

Classes

- class SetOfFDs

5.8 horizon.horizon Namespace Reference

Functions

- None repair_tuple (dict[str, list[str]] columns, int t, FunctionalDependency fd, list[dict[str, str]] repair_table, PatternExpression p_exp, FDPatternGraph fd_pattern_graph)
- tuple[list[PatternExpression], float] repair_dirty_data (Path dirty_data_path, Path cleaned_data_output_↩ path, list[FunctionalDependency] ordered_fds, list[dict[str, str]] repair_table, FDPatternGraph fd_pattern_↩ graph, int|None n_rows=None, bool collect_pattern_expressions=False)
- None run_horizon (str dataset_name, Path dirty_data_path, SetOfFDs set_of_fds, Path output_dir, int|None n_rows=None, bool enable_plotting=False, bool collect_pattern_expressions=False)
- None main (Path dataset_dir, str dirty_data_file, Path output_dir, int|None n_rows, bool enable_plotting, bool collect_pattern_expressions)

Variables

- logging logger = get_logger(__name__)
- argparse parser
- type
- required
- help
- default
- action
- argparse args = parser.parse_args()
- log_level
- exc_info

5.8.1 Function Documentation

5.8.1.1 main()

```
None horizon.horizon.main (
            Path dataset_dir,
            str dirty_data_file,
            Path output_dir,
            int | None n_rows,
            bool enable_plotting,
            bool collect_pattern_expressions )
```

5.8.1.2 repair_dirty_data()

```
tuple[list[PatternExpression], float] horizon.horizon.repair_dirty_data (
            Path dirty_data_path,
            Path cleaned_data_output_path,
            list[FunctionalDependency] ordered_fds,
            list[dict[str, str]] repair_table,
            FDPatternGraph fd_pattern_graph,
            int | None  n_rows = None,
            bool  collect_pattern_expressions = False )
```

```
Repairs data under the given dirty data path, based on the given order and FD Pattern graph.
If n_rows is given, only repairs the first n_rows tuples.
Set collect_pattern_expressions=False to free each tuple's PatternExpression immediately when the lineage is r
Saves cleaned data under given cleaned data output path.
```

5.8.1.3 repair_tuple()

```
None horizon.horizon.repair_tuple (
            dict[str, list[str]] columns,
            int t,
            FunctionalDependency fd,
            list[dict[str, str]] repair_table,
            PatternExpression p_exp,
            FDPatternGraph fd_pattern_graph )
```

```
Given a data frame as a dict, applies in-place repairs for tuple with index t and functional dependency fd.
```

5.8.1.4 run_horizon()

```
None horizon.horizon.run_horizon (
            str dataset_name,
            Path dirty_data_path,
            SetOfFDs set_of_fds,
            Path output_dir,
            int | None  n_rows = None,
            bool  enable_plotting = False,
            bool  collect_pattern_expressions = False )
```

5.8.2 Variable Documentation

5.8.2.1 action

```
horizon.horizon.action
```

5.8.2.2 args

```
argparse horizon.horizon.args = parser.parse_args()
```

5.8.2.3 default

```
horizon.horizon.default
```

5.8.2.4 exc_info

```
horizon.horizon.exc_info
```

5.8.2.5 help

```
horizon.horizon.help
```

5.8.2.6 log_level

```
horizon.horizon.log_level
```

5.8.2.7 logger

```
logging horizon.horizon.logger = get_logger(__name__)
```

5.8.2.8 parser

```
argparse horizon.horizon.parser
```

Initial value:

```
00001 =  argparse.ArgumentParser(
00002     description="Horizon: Scalable Dependency-driven Data Cleaning."
00003 )
```

5.8.2.9 required

```
horizon.horizon.required
```

5.8.2.10 type

```
horizon.horizon.type
```

5.9 horizon.static_fd_analysis Namespace Reference

Classes

- class FDGraph

Functions

- tuple[list[FunctionalDependency], float] get_ordered_fds (SetOfFDs set_of_fds, str dataset_name="", Path output_dir=Path("output"), bool enable_plotting=True)

Variables

- logger = get_logger(__name__)

5.9.1 Function Documentation

5.9.1.1 get_ordered_fds()

```
tuple[list[FunctionalDependency], float] horizon.static_fd_analysis.get_ordered_fds (
            SetOfFDs set_of_fds,
            str  dataset_name = "",
            Path  output_dir = Path("output"),
            bool  enable_plotting = True )
```

5.9.2 Variable Documentation

5.9.2.1 logger

```
horizon.static_fd_analysis.logger = get_logger(__name__)
```

5.10 horizon.utils Namespace Reference

Namespaces

- namespace loaders
- namespace logging_config

Variables

- list __all__ = ["setup_logging", "get_logger"]

5.10.1 Detailed Description

```
Horizon utility modules.
```

5.10.2 Variable Documentation

5.10.2.1 __all__

```
list horizon.utils.__all__ = ["setup_logging", "get_logger"]  [private]
```

5.11 horizon.utils.loaders Namespace Reference

Classes

- class CSVFDLoader
- class FDLoader
- class TXTFDLoader

Functions

- SetOfFDs get_fds (Path source, list[str]|Path columns_or_path)
- SetOfFDs load_fds (Path dataset_dir, Path data_csv_path)
- pl.LazyFrame _scan_csv (Path source, int|None n_rows)
- pl.LazyFrame _scan_parquet (Path source, int|None n_rows)
- pl.DataFrame load_table (Path source, list[str]|None columns=None, int|None n_rows=None)
- pl.LazyFrame lazy_load_table (Path source, list[str]|None columns=None, int|None n_rows=None)
- Iterator[pl.DataFrame] iter_table_batches (Path source, list[str]|None columns=None, int|None n_rows=None, int block_bytes=8 ∗1024 ∗1024)

Variables

- logger = get_logger(__name__)
- dict _SCANNERS

5.11.1 Function Documentation

5.11.1.1 _scan_csv()

```
pl.LazyFrame horizon.utils.loaders._scan_csv (
            Path source,
            int | None n_rows )  [protected]
```

5.11.1.2 _scan_parquet()

```
pl.LazyFrame horizon.utils.loaders._scan_parquet (
            Path source,
            int | None n_rows )  [protected]
```

5.11.1.3 get_fds()

```
SetOfFDs horizon.utils.loaders.get_fds (
            Path source,
            list[str] | Path columns_or_path )
```

```
Extracts FDs as SetOfFDs from the given source. Source can be fds.csv or fds.txt.
```

5.11.1.4 iter_table_batches()

```
Iterator[pl.DataFrame] horizon.utils.loaders.iter_table_batches (
            Path source,
            list[str] | None  columns = None,
            int | None  n_rows = None,
            int  block_bytes = 8 * 1024 * 1024 )
```

```
Yield a table in row batches using memory BOUNDED by block size, not file size.

Additive, non-breaking companion to lazy_load_table for streaming a whole
table once (e.g. the repair loop). For CSV it reads via pyarrow's block-based
reader, whose peak memory is ~O(block_bytes) regardless of file size -- unlike
scan_csv().collect_batches(), which under the polars streaming engine
materialises ~a full copy of the CSV. Columns are read as Utf8 and lowercased,
matching lazy_load_table. Pass 'columns' (any casing) to read only a subset --
projection is pushed into the reader so unselected columns are never parsed;
missing names raise ValueError. Non-CSV sources fall back to the existing lazy
streaming path, so their behaviour is unchanged. Stops after n_rows rows.
```

5.11.1.5 lazy_load_table()

```
pl.LazyFrame horizon.utils.loaders.lazy_load_table (
            Path source,
            list[str] | None  columns = None,
            int | None  n_rows = None )
```

```
Read a CSV or Parquet table with column names lowercased.
Same as load_table() but returns a LazyFrame.

Allows streaming and is more suitable for data larger than RAM.
```

5.11.1.6 load_fds()

```
SetOfFDs horizon.utils.loaders.load_fds (
            Path dataset_dir,
            Path data_csv_path )
```

```
Finds an FD file under the given dataset directory.
Prefers fds.csv, otherwise extracts FDs from fds.txt. Returns SetOfFDs.
```

5.11.1.7 load_table()

```
pl.DataFrame horizon.utils.loaders.load_table (
            Path source,
            list[str] | None  columns = None,
            int | None  n_rows = None )
```

```
Read a CSV or Parquet table with column names lowercased.

Dispatches to a per-extension scanner. Matches the lowercasing in
`CSVFDLoader` so FDs and DataFrame columns share one casing convention.
Pass `columns` (any casing) to read only a subset -- projection is pushed
down into the scan so unselected columns are never read. Missing names
raise ValueError.
```

5.11.2 Variable Documentation

5.11.2.1 _SCANNERS

```
dict horizon.utils.loaders._SCANNERS  [protected]
```

Initial value:

```
00001 =  {
00002     ".csv": _scan_csv,
00003     ".parquet": _scan_parquet,
00004 }
```

5.11.2.2 logger

```
horizon.utils.loaders.logger = get_logger(__name__)
```

5.12 horizon.utils.logging_config Namespace Reference

Functions

- logging.Logger setup_logging (int log_level=logging.INFO, Optional[Path] log_file=None, Path log_dir=Path("logs"))
- logging.Logger get_logger (str name)

5.12.1 Detailed Description

```
Centralized logging configuration for the Horizon pipeline.

Provides consistent logging setup across all modules with:
- File and console logging
- Configurable log levels
- Structured log format with timestamps and module names
```

5.12.2 Function Documentation

5.12.2.1 get_logger()

```
logging.Logger horizon.utils.logging_config.get_logger (
            str name )
```

```
Get a logger for a specific module.

Args:
    name: Module name (typically __name__)

Returns:
    Configured logger instance
```

5.12.2.2 setup_logging()

```
logging.Logger horizon.utils.logging_config.setup_logging (
            int  log_level = logging.INFO,
            Optional[Path]  log_file = None,
            Path  log_dir = Path("logs") )
```

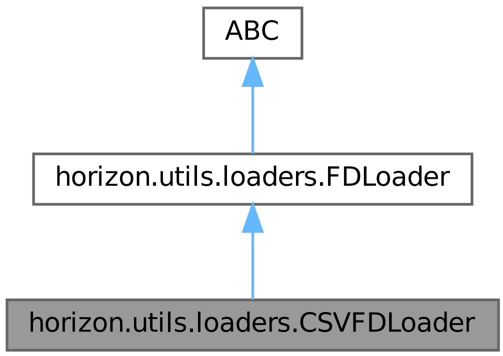
```
Configure logging for the entire Horizon pipeline.

Args:
    log_level: Logging level (logging.DEBUG, logging.INFO, etc.)
    log_file: Path to log file. If None, defaults to logs/horizon.log
    log_dir: Directory to store log files

Returns:
    Configured logger instance
```

Chapter 6

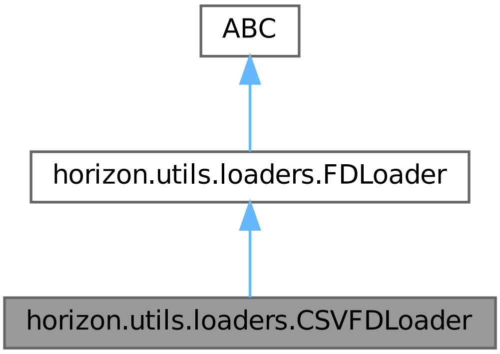
Class Documentation

6.1 horizon.utils.loaders.CSVFDLoader Class Reference

Inheritance diagram for horizon.utils.loaders.CSVFDLoader:



Collaboration diagram for horizon.utils.loaders.CSVFDLoader:



Public Member Functions

- None __init__ (self)
- SetOfFDs load (self, Path source)

6.1.1 Detailed Description

```
Class to load FDs from a CSV file into a SetOfFDs object.
```

6.1.2 Constructor & Destructor Documentation

6.1.2.1 __init__()

```
None horizon.utils.loaders.CSVFDLoader.__init__ (
              self )
```

6.1.3 Member Function Documentation

6.1.3.1 load()

```
SetOfFDs horizon.utils.loaders.CSVFDLoader.load (
              self,
             Path source )
```

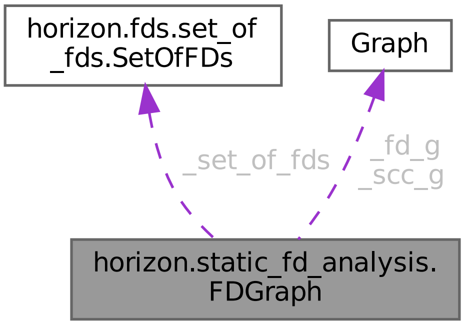
Reimplemented from horizon.utils.loaders.FDLoader.

The documentation for this class was generated from the following file:

- horizon/utils/loaders.py

6.2 horizon.static_fd_analysis.FDGraph Class Reference

Collaboration diagram for horizon.static_fd_analysis.FDGraph:



Public Member Functions

- None __init__ (self, SetOfFDs set_of_fds)
- list[FunctionalDependency] order_fds (self)
- list[int] dfs_tree (self, str start_attribute)
- None plot_graphs (self, str dataset_name, Path output_dir)
- dict graph_data (self)

Protected Member Functions

- Graph _build_fd_graph (self)
- Graph _find_strongly_connected_components (self)
- list[FunctionalDependency] _order_subg (self, Graph sub_g, int start_v, int order_counter)
- list[FunctionalDependency] _get_topological_sorting (self, Graph g, int order_counter)

Protected Attributes

- _set_of_fds
- _fd_g
- _scc_g
- _components

Static Protected Attributes

- SetOfFDs _set_of_fds
- Graph _fd_g
- Graph _scc_g
- ig _components .VertexClustering

6.2.1 Constructor & Destructor Documentation

6.2.1.1 __init__()

```
None horizon.static_fd_analysis.FDGraph.__init__ (
            self,
            SetOfFDs set_of_fds )
```

6.2.2 Member Function Documentation

6.2.2.1 _build_fd_graph()

```
Graph horizon.static_fd_analysis.FDGraph._build_fd_graph (
            self )  [protected]
```

6.2.2.2 _find_strongly_connected_components()

```
Graph horizon.static_fd_analysis.FDGraph._find_strongly_connected_components (
            self )  [protected]
```

```
Finds strongly connected components and builds SCC graph.
```

6.2.2.3 _get_topological_sorting()

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph._get_topological_sorting (
            self,
            Graph g,
            int  order_counter )  [protected]
```

```
Implements Kahn's Algorithm for computing a topological sorting using BFS.
```

6.2.2.4 _order_subg()

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph._order_subg (
            self,
            Graph sub_g,
            int start_v,
            int  order_counter )  [protected]
```

```
Implements Hierholzer's linear time algorithm for constructing an Eulerian cycle/tour.
```

6.2.2.5 dfs_tree()

```
list[int] horizon.static_fd_analysis.FDGraph.dfs_tree (
              self,
             str start_attribute )
```

Constructs a DFS tree and returns all visited edges.

6.2.2.6 graph_data()

```
dict horizon.static_fd_analysis.FDGraph.graph_data (
              self )
```

Structured, JSON-serialisable description of the graphs for the UI.

Takes computed FD graph, SCCG, and FD order data and stores it as JSON data.

6.2.2.7 order_fds()

```
list[FunctionalDependency] horizon.static_fd_analysis.FDGraph.order_fds (
              self )
```

Orders each independent sub-component of the FD graph.

6.2.2.8 plot_graphs()

```
None horizon.static_fd_analysis.FDGraph.plot_graphs (
              self,
             str dataset_name,
             Path output_dir )
```

6.2.3 Member Data Documentation

6.2.3.1 _components [1/2]

```
ig horizon.static_fd_analysis.FDGraph._components .VertexClustering  [static], [protected]
```

6.2.3.2 _components [2/2]

```
horizon.static_fd_analysis.FDGraph._components  [protected]
```

6.2.3.3 _fd_g [1/2]

```
Graph horizon.static_fd_analysis.FDGraph._fd_g  [static], [protected]
```

6.2.3.4 _fd_g [2/2]

`horizon.static_fd_analysis.FDGraph._fd_g [protected]`

6.2.3.5 _scc_g [1/2]

`Graph horizon.static_fd_analysis.FDGraph._scc_g [static], [protected]`

6.2.3.6 _scc_g [2/2]

`horizon.static_fd_analysis.FDGraph._scc_g [protected]`

6.2.3.7 _set_of_fds [1/2]

`SetOfFDs horizon.static_fd_analysis.FDGraph._set_of_fds [static], [protected]`

6.2.3.8 _set_of_fds [2/2]

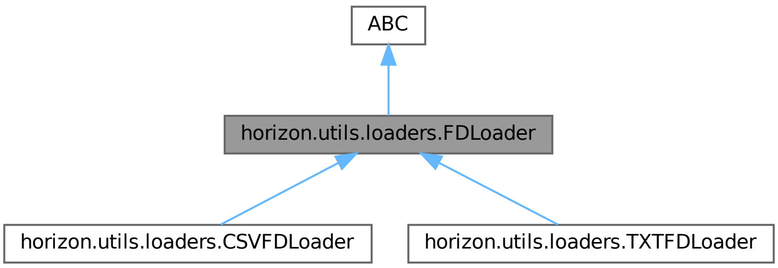
`horizon.static_fd_analysis.FDGraph._set_of_fds [protected]`

The documentation for this class was generated from the following file:

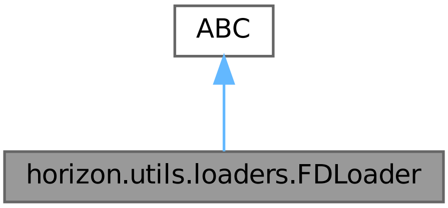
- horizon/static_fd_analysis.py

6.3 horizon.utils.loaders.FDLoader Class Reference

Inheritance diagram for horizon.utils.loaders.FDLoader:



Collaboration diagram for horizon.utils.loaders.FDLoader:



Public Member Functions

- SetOfFDs load (self, source)

6.3.1 Detailed Description

```
Abstract base class for FD loaders.
```

6.3.2 Member Function Documentation

6.3.2.1 load()

```
SetOfFDs horizon.utils.loaders.FDLoader.load (
              self,
              source )
```

Reimplemented in horizon.utils.loaders.CSVFDLoader, and horizon.utils.loaders.TXTFDLoader.

The documentation for this class was generated from the following file:

- horizon/utils/loaders.py

6.4 horizon.fds.fd_pattern.FDPattern Class Reference

Public Member Functions

- None __init__ (self, FunctionalDependency fd, tuple|list|str lval, str rval)
- FunctionalDependency fd (self)
- tuple lval (self)
- str rval (self)
- str __repr__ (self)
- bool __eq__ (self, object other)
- int __hash__ (self)

Protected Attributes

- _fd
- _lval
- _rval

6.4.1 Constructor & Destructor Documentation

6.4.1.1 __init__()

```
None horizon.fds.fd_pattern.FDPattern.__init__ (
             self,
            FunctionalDependency fd,
            tuple | list | str lval,
            str  rval )
```

6.4.2 Member Function Documentation

6.4.2.1 __eq__()

```
bool horizon.fds.fd_pattern.FDPattern.__eq__ (
             self,
            object other )
```

6.4.2.2 __hash__()

```
int horizon.fds.fd_pattern.FDPattern.__hash__ (
             self )
```

6.4.2.3 __repr__()

```
str horizon.fds.fd_pattern.FDPattern.__repr__ (
             self )
```

6.4.2.4 fd()

```
FunctionalDependency horizon.fds.fd_pattern.FDPattern.fd (
             self )
```

6.4.2.5 lval()

```
tuple horizon.fds.fd_pattern.FDPattern.lval (
             self )
```

6.4.2.6 rval()

```
str horizon.fds.fd_pattern.FDPattern.rval (
              self )
```

6.4.3 Member Data Documentation

6.4.3.1 _fd

```
horizon.fds.fd_pattern.FDPattern._fd  [protected]
```

6.4.3.2 _lval

```
horizon.fds.fd_pattern.FDPattern._lval  [protected]
```

6.4.3.3 _rval

```
horizon.fds.fd_pattern.FDPattern._rval  [protected]
```

The documentation for this class was generated from the following file:

- horizon/fds/fd_pattern.py

6.5 horizon.fd_pattern_graph.FDPatternGraph Class Reference

Public Member Functions

- None __init__ (self, Path data_path, SetOfFDs set_of_fds, str dataset_name="", bool pruning=True, Path output_dir=Path("output"), bool enable_plotting=False)
- list[dict[str, str]] repair_table (self)
- int number_of_nodes (self)
- int number_of_edges (self)
- float get_edge_quality (self, str from_node, str to_node)
- dict get_node_info (self, str node_id)
- bool has_node (self, str col_name, str value)
- tuple[str, str] choose_best_next_edge (self, str lhs, str lval, str rhs)
- None clear (self)

Public Attributes

- graph

Static Public Attributes

- nx graph .DiGraph

Protected Member Functions

- str _cell_node_id (self, str col_name, value)
- tuple[str, str] _parse_node_id (self, str node_id)
- nx.DiGraph _build_graph (self, Path data_path, SetOfFDs set_of_fds)
- None _prune_edges (self)

Protected Attributes

- _pruning
- _repair_table

Static Protected Attributes

- list _repair_table [dict[str, str]]

6.5.1 Detailed Description

```
A graph representation of CSV data with functional dependency edges.

Nodes represent unique (column, value) pairs. Multiple cells with the
same value in the same column will reference the same node.
```

6.5.2 Constructor & Destructor Documentation

6.5.2.1 __init__()

```
None horizon.fd_pattern_graph.FDPatternGraph.__init__ (
              self,
             Path data_path,
             SetOfFDs set_of_fds,
             str  dataset_name = "",
             bool  pruning = True,
             Path  output_dir = Path("output"),
             bool  enable_plotting = False )
```

```
Initialize the FDPatternGraph.

Args:
    data_path: Path to the CSV file containing the data
    set_of_fds: Parsed set of functional dependencies
```

6.5.3 Member Function Documentation

6.5.3.1 _build_graph()

```
nx.DiGraph horizon.fd_pattern_graph.FDPatternGraph._build_graph (
              self,
             Path data_path,
             SetOfFDs set_of_fds )  [protected]
```

```
Build the graph with nodes for unique (column, value) pairs
and edges based on functional dependencies.
Calculate quality for each edge in the graph and store as edge attributes.

Returns:
    NetworkX DiGraph with column-value nodes and FD-based edges
```

6.5.3.2 _cell_node_id()

```
str horizon.fd_pattern_graph.FDPatternGraph._cell_node_id (
              self,
             str col_name,
              value )  [protected]
```

```
Generate a unique node ID for a (column, value) pair.

Format: "{col_name}__{value}" - e.g., "brewery-name__Guinness"
Same value in the same column always produces the same node ID.
Different columns get different nodes even for the same value.
```

6.5.3.3 _parse_node_id()

```
tuple[str, str] horizon.fd_pattern_graph.FDPatternGraph._parse_node_id (
              self,
             str node_id )  [protected]
```

```
Parse a node ID back to (col_name, value).

Args:
    node_id: Node identifier like "brewery-name_Guinness"

Returns:
    Tuple of (column_name, value)
```

6.5.3.4 _prune_edges()

```
None horizon.fd_pattern_graph.FDPatternGraph._prune_edges (
              self )  [protected]
```

6.5.3.5 choose_best_next_edge()

```
tuple[str, str] horizon.fd_pattern_graph.FDPatternGraph.choose_best_next_edge (
              self,
             str lhs,
             str lval,
             str rhs )
```

```
Choose best next edge given a LHS attribute and value, returns RHS attribute and value with the highest qualit
```

6.5.3.6 clear()

```
None horizon.fd_pattern_graph.FDPatternGraph.clear (
              self )
```

```
Clears the FD Pattern graph.
```

6.5.3.7 get_edge_quality()

```
float horizon.fd_pattern_graph.FDPatternGraph.get_edge_quality (
              self,
             str from_node,
             str to_node )
```

```
Get the quality of a specific edge.

Args:
    from_node: Source node ID
    to_node: Target node ID

Returns:
    Quality score for the edge
```

6.5.3.8 get_node_info()

```
dict horizon.fd_pattern_graph.FDPatternGraph.get_node_info (
              self,
             str node_id )
```

```
Get all stored information about a node.
```

6.5.3.9 has_node()

```
bool horizon.fd_pattern_graph.FDPatternGraph.has_node (
              self,
             str col_name,
             str value )
```

```
True if a (column, value) node exists in the graph.
```

6.5.3.10 number_of_edges()

```
int horizon.fd_pattern_graph.FDPatternGraph.number_of_edges (
              self )
```

6.5.3.11 number_of_nodes()

```
int horizon.fd_pattern_graph.FDPatternGraph.number_of_nodes (
              self )
```

6.5.3.12 repair_table()

```
list[dict[str, str]] horizon.fd_pattern_graph.FDPatternGraph.repair_table (
              self )
```

6.5.4 Member Data Documentation

6.5.4.1 _pruning

`horizon.fd_pattern_graph.FDPatternGraph._pruning [protected]`

6.5.4.2 _repair_table [1/2]

`list horizon.fd_pattern_graph.FDPatternGraph._repair_table [dict[str, str]] [static], [protected]`

6.5.4.3 _repair_table [2/2]

`horizon.fd_pattern_graph.FDPatternGraph._repair_table [protected]`

6.5.4.4 graph [1/2]

`nx horizon.fd_pattern_graph.FDPatternGraph.graph .DiGraph [static]`

6.5.4.5 graph [2/2]

`horizon.fd_pattern_graph.FDPatternGraph.graph`

The documentation for this class was generated from the following file:

- horizon/fd_pattern_graph.py

6.6 horizon.fds.fd.FunctionalDependency Class Reference

Public Member Functions

- None __init__ (self, tuple|list|str lhs, str rhs, int index=0, int|None order=None)
- str|tuple lhs (self)
- tuple lhs_attributes (self)
- str rhs (self)
- int index (self)
- None index (self, int index)
- int|None order (self)
- None order (self, int order)
- bool|None cyclic (self)
- None cyclic (self, bool cyclic)
- str __repr__ (self)
- bool __eq__ (self, object other)
- int __hash__ (self)
- tuple as_tuple (self)
- list[str] get_attributes (self)

Public Attributes

- lhs

Protected Attributes

- _lhs
- _rhs
- _index
- _order
- _cyclic

Static Protected Attributes

- int _index
- int _order | None
- bool _cyclic | None

6.6.1 Constructor & Destructor Documentation

6.6.1.1 __init__()

```
None horizon.fds.fd.FunctionalDependency.__init__ (
              self,
             tuple | list | str lhs,
             str rhs,
             int  index = 0,
             int | None  order = None )
```

6.6.2 Member Function Documentation

6.6.2.1 __eq__()

```
bool horizon.fds.fd.FunctionalDependency.__eq__ (
              self,
             object other )
```

6.6.2.2 __hash__()

```
int horizon.fds.fd.FunctionalDependency.__hash__ (
              self )
```

6.6.2.3 __repr__()

```
str horizon.fds.fd.FunctionalDependency.__repr__ (
              self )
```

6.6.2.4 as_tuple()

```
tuple horizon.fds.fd.FunctionalDependency.as_tuple (
              self )
```

6.6.2.5 cyclic() [1/2]

```
bool | None horizon.fds.fd.FunctionalDependency.cyclic (
              self )
```

6.6.2.6 cyclic() [2/2]

```
None horizon.fds.fd.FunctionalDependency.cyclic (
              self,
             bool cyclic )
```

6.6.2.7 get_attributes()

```
list[str] horizon.fds.fd.FunctionalDependency.get_attributes (
              self )
```

6.6.2.8 index() [1/2]

```
int horizon.fds.fd.FunctionalDependency.index (
              self )
```

6.6.2.9 index() [2/2]

```
None horizon.fds.fd.FunctionalDependency.index (
              self,
             int index )
```

6.6.2.10 lhs()

```
str | tuple horizon.fds.fd.FunctionalDependency.lhs (
              self )
```

6.6.2.11 lhs_attributes()

```
tuple horizon.fds.fd.FunctionalDependency.lhs_attributes (
              self )
```

6.6.2.12 order() [1/2]

```
int | None horizon.fds.fd.FunctionalDependency.order (
              self )
```

6.6.2.13 order() [2/2]

```
None horizon.fds.fd.FunctionalDependency.order (
              self,
             int order )
```

6.6.2.14 rhs()

```
str horizon.fds.fd.FunctionalDependency.rhs (
              self )
```

6.6.3 Member Data Documentation

6.6.3.1 _cyclic [1/2]

```
bool horizon.fds.fd.FunctionalDependency._cyclic | None  [static], [protected]
```

6.6.3.2 _cyclic [2/2]

```
horizon.fds.fd.FunctionalDependency._cyclic  [protected]
```

6.6.3.3 _index [1/2]

```
int horizon.fds.fd.FunctionalDependency._index  [static], [protected]
```

6.6.3.4 _index [2/2]

```
horizon.fds.fd.FunctionalDependency._index  [protected]
```

6.6.3.5 _lhs

```
horizon.fds.fd.FunctionalDependency._lhs  [protected]
```

6.6.3.6 _order [1/2]

```
int horizon.fds.fd.FunctionalDependency._order | None  [static], [protected]
```

6.6.3.7 _order [2/2]

```
horizon.fds.fd.FunctionalDependency._order [protected]
```

6.6.3.8 _rhs

```
horizon.fds.fd.FunctionalDependency._rhs [protected]
```

6.6.3.9 lhs

```
horizon.fds.fd.FunctionalDependency.lhs
```

The documentation for this class was generated from the following file:

- horizon/fds/fd.py

6.7 horizon.fds.pattern_expression.PatternExpression Class Reference

Public Member Functions

- None __init__ (self, int tuple_index)
- int tuple_index (self)
- str __repr__ (self)
- bool __eq__ (self, object other)
- int __hash__ (self)
- FDPattern __getitem__ (self, int index)
- Iterator[FDPattern] __iter__ (self)
- int __len__ (self)
- None add_fd_pattern (self, FDPattern fd_pattern)
- FDPattern|None attribute_in_expression (self, str attribute)

Protected Attributes

- _fd_patterns

Static Protected Attributes

- list _fd_patterns [FDPattern]

6.7.1 Constructor & Destructor Documentation

6.7.1.1 __init__()

```
None horizon.fds.pattern_expression.PatternExpression.__init__ (
             self,
            int tuple_index )
```

6.7.2 Member Function Documentation

6.7.2.1 __eq__()

```
bool horizon.fds.pattern_expression.PatternExpression.__eq__ (
            self,
            object other )
```

6.7.2.2 __getitem__()

```
FDPattern horizon.fds.pattern_expression.PatternExpression.__getitem__ (
            self,
            int index )
```

6.7.2.3 __hash__()

```
int horizon.fds.pattern_expression.PatternExpression.__hash__ (
            self )
```

6.7.2.4 __iter__()

```
Iterator[FDPattern] horizon.fds.pattern_expression.PatternExpression.__iter__ (
            self )
```

6.7.2.5 __len__()

```
int horizon.fds.pattern_expression.PatternExpression.__len__ (
            self )
```

6.7.2.6 __repr__()

```
str horizon.fds.pattern_expression.PatternExpression.__repr__ (
            self )
```

6.7.2.7 add_fd_pattern()

```
None horizon.fds.pattern_expression.PatternExpression.add_fd_pattern (
            self,
            FDPattern fd_pattern )
```

6.7.2.8 attribute_in_expression()

```
FDPattern | None horizon.fds.pattern_expression.PatternExpression.attribute_in_expression (
            self,
            str attribute )
```

6.7.2.9 tuple_index()

```
int horizon.fds.pattern_expression.PatternExpression.tuple_index (
              self )
```

6.7.3 Member Data Documentation

6.7.3.1 _fd_patterns [1/2]

```
list horizon.fds.pattern_expression.PatternExpression._fd_patterns [FDPattern]  [static],
[protected]
```

6.7.3.2 _fd_patterns [2/2]

```
horizon.fds.pattern_expression.PatternExpression._fd_patterns  [protected]
```

The documentation for this class was generated from the following file:

- horizon/fds/pattern_expression.py

6.8 horizon.fds.set_of_fds.SetOfFDs Class Reference

Public Member Functions

- None __init__ (self, list[FunctionalDependency]|None set_of_fds, set[str]|None bound_attributes=None)
- list[FunctionalDependency] set_of_fds (self)
- set[str] unique_attributes (self)
- set[str] bound_attributes (self)
- None bound_attributes (self, set[str] bound_attributes)
- set[str] source_attributes (self)
- None source_attributes (self, set[str] source_attributes)
- list[FunctionalDependency] get_ordered_set_of_fds (self)
- None add_fd (self, FunctionalDependency fd)
- list[tuple] as_tuple_list (self)
- bool is_bound (self, str attribute)
- bool is_source (self, str attribute)
- str __repr__ (self)
- bool __eq__ (self, object other)
- int __hash__ (self)
- FunctionalDependency __getitem__ (self, int index)
- Iterator[FunctionalDependency] __iter__ (self)
- int __len__ (self)

Protected Attributes

- _set_of_fds
- _bound_attributes
- _source_attributes
- _unique_attributes

Static Protected Attributes

- list _set_of_fds [FunctionalDependency]
- set _unique_attributes [str]
- set _bound_attributes [str]
- set _source_attributes [str]

6.8.1 Constructor & Destructor Documentation

6.8.1.1 __init__()

```
None horizon.fds.set_of_fds.SetOfFDs.__init__ (
             self,
             list[FunctionalDependency] | None set_of_fds,
             set[str] | None bound_attributes = None )
```

6.8.2 Member Function Documentation

6.8.2.1 __eq__()

```
bool horizon.fds.set_of_fds.SetOfFDs.__eq__ (
             self,
             object other )
```

6.8.2.2 __getitem__()

```
FunctionalDependency horizon.fds.set_of_fds.SetOfFDs.__getitem__ (
             self,
             int index )
```

6.8.2.3 __hash__()

```
int horizon.fds.set_of_fds.SetOfFDs.__hash__ (
             self )
```

6.8.2.4 __iter__()

```
Iterator[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.__iter__ (
             self )
```

6.8.2.5 __len__()

```
int horizon.fds.set_of_fds.SetOfFDs.__len__ (
             self )
```

6.8.2.6 __repr__()

```
str horizon.fds.set_of_fds.SetOfFDs.__repr__ (
              self )
```

6.8.2.7 add_fd()

```
None horizon.fds.set_of_fds.SetOfFDs.add_fd (
              self,
             FunctionalDependency fd )
```

```
Adds a new FD, adjusting its index (to preserve uniqueness),
and updating unique attributes.
```

6.8.2.8 as_tuple_list()

```
list[tuple] horizon.fds.set_of_fds.SetOfFDs.as_tuple_list (
              self )
```

```
Returns the set of FDs as tuples
```

6.8.2.9 bound_attributes() [1/2]

```
set[str] horizon.fds.set_of_fds.SetOfFDs.bound_attributes (
              self )
```

6.8.2.10 bound_attributes() [2/2]

```
None horizon.fds.set_of_fds.SetOfFDs.bound_attributes (
              self,
             set[str] bound_attributes )
```

6.8.2.11 get_ordered_set_of_fds()

```
list[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.get_ordered_set_of_fds (
              self )
```

```
Returns ordered set of FDs, set by static_fd_analyis.py.
```

6.8.2.12 is_bound()

```
bool horizon.fds.set_of_fds.SetOfFDs.is_bound (
              self,
             str attribute )
```

```
True if attribute is bound.
```

6.8.2.13 is_source()

```
bool horizon.fds.set_of_fds.SetOfFDs.is_source (
             self,
            str attribute )
```

```
True if attribute never appears as an FD RHS (a source column).
```

6.8.2.14 set_of_fds()

```
list[FunctionalDependency] horizon.fds.set_of_fds.SetOfFDs.set_of_fds (
             self )
```

6.8.2.15 source_attributes() [1/2]

```
set[str] horizon.fds.set_of_fds.SetOfFDs.source_attributes (
             self )
```

6.8.2.16 source_attributes() [2/2]

```
None horizon.fds.set_of_fds.SetOfFDs.source_attributes (
             self,
            set[str] source_attributes )
```

6.8.2.17 unique_attributes()

```
set[str] horizon.fds.set_of_fds.SetOfFDs.unique_attributes (
             self )
```

6.8.3 Member Data Documentation

6.8.3.1 _bound_attributes [1/2]

```
set horizon.fds.set_of_fds.SetOfFDs._bound_attributes [str]  [static], [protected]
```

6.8.3.2 _bound_attributes [2/2]

```
horizon.fds.set_of_fds.SetOfFDs._bound_attributes  [protected]
```

6.8.3.3 _set_of_fds [1/2]

```
list horizon.fds.set_of_fds.SetOfFDs._set_of_fds [FunctionalDependency]  [static], [protected]
```

6.8.3.4 _set_of_fds [2/2]

```
horizon.fds.set_of_fds.SetOfFDs._set_of_fds  [protected]
```

6.8.3.5 _source_attributes [1/2]

```
set horizon.fds.set_of_fds.SetOfFDs._source_attributes [str]  [static], [protected]
```

6.8.3.6 _source_attributes [2/2]

```
horizon.fds.set_of_fds.SetOfFDs._source_attributes  [protected]
```

6.8.3.7 _unique_attributes [1/2]

```
set horizon.fds.set_of_fds.SetOfFDs._unique_attributes [str]  [static], [protected]
```

6.8.3.8 _unique_attributes [2/2]

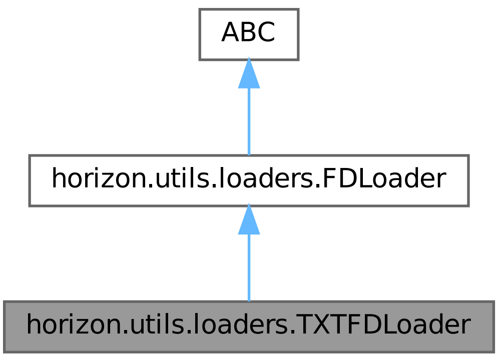
```
horizon.fds.set_of_fds.SetOfFDs._unique_attributes  [protected]
```

The documentation for this class was generated from the following file:

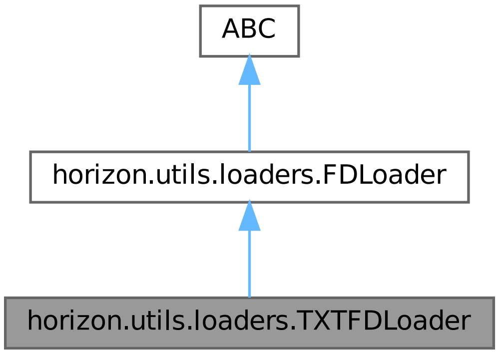
- horizon/fds/set_of_fds.py

6.9 horizon.utils.loaders.TXTFDLoader Class Reference

Inheritance diagram for horizon.utils.loaders.TXTFDLoader:



Collaboration diagram for horizon.utils.loaders.TXTFDLoader:



Public Member Functions

- None __init__ (self, list[str]|Path columns_or_path, str separator="->")
- list[str] load_column_names (self, Path columns_csv_path)
- tuple[list[str], str]|None parse_line (self, line)
- SetOfFDs load (self, Path source)

Protected Attributes

- _column_names

Static Protected Attributes

- dict _EXTENSION_LOADERS

6.9.1 Detailed Description

```
Class to load FDs from a TXT file into a SetOfFDs object. Assumes FDs in the file
are represented by column indices and a separator (e.g., 1 -> 5).
```

6.9.2 Constructor & Destructor Documentation

6.9.2.1 __init__()

```
None horizon.utils.loaders.TXTFDLoader.__init__ (
              self,
             list[str] | Path columns_or_path,
             str  separator = "->" )
```

```
Constructor requires a list of columns or a CSV path from which the column
names can be extracted. It also requires a separator (default: ->).
```

6.9.3 Member Function Documentation

6.9.3.1 load()

```
SetOfFDs horizon.utils.loaders.TXTFDLoader.load (
             self,
            Path source )
```

Reimplemented from horizon.utils.loaders.FDLoader.

6.9.3.2 load_column_names()

```
list[str] horizon.utils.loaders.TXTFDLoader.load_column_names (
             self,
            Path columns_csv_path )
```

```
Loads column names from a given CSV file and returns them as a list.
```

6.9.3.3 parse_line()

```
tuple[list[str], str] | None horizon.utils.loaders.TXTFDLoader.parse_line (
             self,
             line )
```

```
Helper function to parse one line of a file and translate it into
a tuple containing the FD's LHS and RHS.
```

6.9.4 Member Data Documentation

6.9.4.1 _column_names

```
horizon.utils.loaders.TXTFDLoader._column_names  [protected]
```

6.9.4.2 _EXTENSION_LOADERS

```
dict horizon.utils.loaders.TXTFDLoader._EXTENSION_LOADERS  [static], [protected]
```

Initial value:

```
=  {
    ".csv": CSVFDLoader,
    ".txt": TXTFDLoader,
}
```

The documentation for this class was generated from the following file:

- horizon/utils/loaders.py

Chapter 7

File Documentation

7.1 horizon/fd_pattern_graph.py File Reference

Classes

- class horizon.fd_pattern_graph.FDPatternGraph

Namespaces

- namespace horizon
- namespace horizon.fd_pattern_graph

Variables

- horizon.fd_pattern_graph.logger = get_logger(__name__)

7.2 horizon/__init__.py File Reference

7.3 horizon/fds/__init__.py File Reference

Namespaces

- namespace horizon
- namespace horizon.fds

7.4 horizon/utils/__init__.py File Reference

Namespaces

- namespace horizon
- namespace horizon.utils

Variables

- list horizon.utils.__all__ = ["setup_logging", "get_logger"]

7.5 horizon/fds/fd.py File Reference

Classes

- class horizon.fds.fd.FunctionalDependency

Namespaces

- namespace horizon
- namespace horizon.fds
- namespace horizon.fds.fd

7.6 horizon/fds/fd_pattern.py File Reference

Classes

- class horizon.fds.fd_pattern.FDPattern

Namespaces

- namespace horizon
- namespace horizon.fds
- namespace horizon.fds.fd_pattern

7.7 horizon/fds/pattern_expression.py File Reference

Classes

- class horizon.fds.pattern_expression.PatternExpression

Namespaces

- namespace horizon
- namespace horizon.fds
- namespace horizon.fds.pattern_expression

7.8 horizon/fds/set_of_fds.py File Reference

Classes

- class horizon.fds.set_of_fds.SetOfFDs

Namespaces

- namespace horizon
- namespace horizon.fds
- namespace horizon.fds.set_of_fds

7.9 horizon/horizon.py File Reference

Namespaces

- namespace horizon
- namespace horizon.horizon

Functions

- None horizon.horizon.repair_tuple (dict[str, list[str]] columns, int t, FunctionalDependency fd, list[dict[str, str]] repair_table, PatternExpression p_exp, FDPatternGraph fd_pattern_graph)
- tuple[list[PatternExpression], float] horizon.horizon.repair_dirty_data (Path dirty_data_path, Path cleaned↩ _data_output_path, list[FunctionalDependency] ordered_fds, list[dict[str, str]] repair_table, FDPatternGraph fd_pattern_graph, int|None n_rows=None, bool collect_pattern_expressions=False)
- None horizon.horizon.run_horizon (str dataset_name, Path dirty_data_path, SetOfFDs set_of_fds, Path output_dir, int|None n_rows=None, bool enable_plotting=False, bool collect_pattern_expressions=False)
- None horizon.horizon.main (Path dataset_dir, str dirty_data_file, Path output_dir, int|None n_rows, bool enable_plotting, bool collect_pattern_expressions)

Variables

- logging horizon.horizon.logger = get_logger(__name__)
- argparse horizon.horizon.parser
- horizon.horizon.type
- horizon.horizon.required
- horizon.horizon.help
- horizon.horizon.default
- horizon.horizon.action
- argparse horizon.horizon.args = parser.parse_args()
- horizon.horizon.log_level
- horizon.horizon.exc_info

7.10 horizon/static_fd_analysis.py File Reference

Classes

- class horizon.static_fd_analysis.FDGraph

Namespaces

- namespace horizon
- namespace horizon.static_fd_analysis

Functions

- tuple[list[FunctionalDependency], float] horizon.static_fd_analysis.get_ordered_fds (SetOfFDs set_of_fds, str dataset_name="", Path output_dir=Path("output"), bool enable_plotting=True)

Variables

- horizon.static_fd_analysis.logger = get_logger(__name__)

7.11 horizon/utils/loaders.py File Reference

Classes

- class horizon.utils.loaders.FDLoader
- class horizon.utils.loaders.CSVFDLoader
- class horizon.utils.loaders.TXTFDLoader

Namespaces

- namespace horizon
- namespace horizon.utils
- namespace horizon.utils.loaders

Functions

- SetOfFDs horizon.utils.loaders.get_fds (Path source, list[str]|Path columns_or_path)
- SetOfFDs horizon.utils.loaders.load_fds (Path dataset_dir, Path data_csv_path)
- pl.LazyFrame horizon.utils.loaders._scan_csv (Path source, int|None n_rows)
- pl.LazyFrame horizon.utils.loaders._scan_parquet (Path source, int|None n_rows)
- pl.DataFrame horizon.utils.loaders.load_table (Path source, list[str]|None columns=None, int|None n_↩ rows=None)
- pl.LazyFrame horizon.utils.loaders.lazy_load_table (Path source, list[str]|None columns=None, int|None n↩ _rows=None)
- Iterator[pl.DataFrame] horizon.utils.loaders.iter_table_batches (Path source, list[str]|None columns=None, int|None n_rows=None, int block_bytes=8 ∗1024 ∗1024)

Variables

- horizon.utils.loaders.logger = get_logger(__name__)
- dict horizon.utils.loaders._SCANNERS

7.12 horizon/utils/logging_config.py File Reference

Namespaces

- namespace horizon
- namespace horizon.utils
- namespace horizon.utils.logging_config

Functions

- logging.Logger horizon.utils.logging_config.setup_logging (int log_level=logging.INFO, Optional[Path] log_↩ file=None, Path log_dir=Path("logs"))
- logging.Logger horizon.utils.logging_config.get_logger (str name)

Index